Supporting Information

Accept or abstain? A pre-specified test of reliability gates calibrated on held-out chemistry for molecular property and reaction-yield models

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S1 Results of the MLP ensemble (M2)

Table S1. Ungated accuracy of M2 (mean ± s.d. over five seeds). Error rate: share of predictions with an absolute error above 0.5 log units (logP) or 10 percentage points (yield).

Test set	n	RMSE	R^2	Error rate
LogP, random R	2,449	0.62 ± 0.01	0.87 ± 0.01	0.30
LogP, held-out cluster	3,551	0.84 ± 0.02	0.74 ± 0.02	0.45
LogP, prospective E1	327	0.63 ± 0.03	0.61 ± 0.03	0.41
LogP, fluorinated E2 (all)	183	0.70 ± 0.02	-0.42 ± 0.07	0.48
sp^3 C–F compounds	140	0.70 ± 0.03	-0.70 ± 0.12	0.50
parents	43	0.70 ± 0.02	0.04 ± 0.06	0.42
BH Test1, random R	350	6.19 ± 0.45	0.95 ± 0.01	0.09
BH Test1, held-out additives	898	12.08 ± 0.95	0.80 ± 0.03	0.30
BH Test2, random R	351	6.15 ± 0.46	0.95 ± 0.00	0.10
BH Test2, held-out additives	900	12.69 ± 0.55	0.78 ± 0.02	0.34
BH Test3, random R	350	5.52 ± 0.36	0.96 ± 0.01	0.08
BH Test3, held-out additives	897	16.20 ± 0.99	0.67 ± 0.04	0.40
BH Test4, random R	350	5.47 ± 0.44	0.96 ± 0.01	0.08
BH Test4, held-out additives	900	17.87 ± 0.93	0.54 ± 0.05	0.48
SM, random R	576	10.91 ± 0.69	0.85 ± 0.02	0.25
SM, held-out ligands	960	19.48 ± 4.59	0.28 ± 0.61	0.45

Table S2. AURC of the selective error rate for M2 (lower is better; mean over seeds). "No gate" is the ungated error rate.

Held-out test	No gate	s_1	$-u$	Fused
LogP cluster	0.445	0.339	0.366	0.314
LogP prospective (E1)	0.408	0.292	0.373	0.284
LogP fluorinated (E2)	0.480	0.416	0.434	0.392
BH Test1	0.304	0.280	0.246	0.246
BH Test2	0.336	0.352	0.259	0.259
BH Test3	0.396	0.351	0.337	0.339
BH Test4	0.481	0.483	0.446	0.461
SM ligand	0.454	0.463	0.360	0.386

Table S3. Operating-limits card for Gate B with M2 at $t = 0.8$. Pooled precision with Wilson 95% intervals; other definitions as in Table 4 of the main text. The C_h-calibrated gate abstained on every query in every seed.

	Calibrated on C_r		Calibrated on C_h		
Held-out test	Coverage	Precision [95% CI]	Coverage	Precision [95% CI]	Seeds accepting
LogP cluster	0.07	0.82 [0.80, 0.84]	0.00	abstained	0/5
LogP prospective (E1)	0.08	0.81 [0.74, 0.87]	0.00	abstained	0/5
LogP fluorinated (E2)	0.06	0.69 [0.56, 0.80]	0.00	abstained	0/5
BH Test1	0.98	0.70 [0.69, 0.72]	0.00	abstained	0/5
BH Test2	0.97	0.67 [0.66, 0.69]	0.00	abstained	0/5
BH Test3	0.79	0.60 [0.59, 0.62]	0.00	abstained	0/5
BH Test4	0.99	0.52 [0.51, 0.54]	0.00	abstained	0/5
SM ligand	0.13	0.60 [0.56, 0.64]	0.00	abstained	0/5

S2 Operating-limits card of M1 at $t = 0.9$

Table S4. Operating-limits card for Gate B with M1 at $t = 0.9$. Definitions as in Table 4 of the main text.

	Calibrated on C_r		Calibrated on C_h		
Held-out test	Coverage	Precision [95% CI]	Coverage	Precision [95% CI]	Seeds accepting
LogP cluster	0.01	0.91 [0.84, 0.95]	0.00	0.92 [0.83, 0.97]	1/5
LogP prospective (E1)	0.01	0.92 [0.67, 0.99]	0.01	1.00 [0.72, 1.00]	1/5
LogP fluorinated (E2)	0.00	abstained	0.00	1.00 [0.34, 1.00]	1/5
BH Test1	0.45	0.91 [0.89, 0.92]	0.18	0.94 [0.93, 0.96]	3/5
BH Test2	0.61	0.87 [0.86, 0.88]	0.22	0.95 [0.94, 0.96]	4/5
BH Test3	0.34	0.85 [0.83, 0.87]	0.10	0.95 [0.92, 0.96]	3/5
BH Test4	0.70	0.65 [0.63, 0.66]	0.19	0.72 [0.69, 0.75]	2/5
SM ligand	0.27	0.88 [0.86, 0.90]	0.04	0.90 [0.85, 0.93]	1/5

S3 Analysis protocol as committed

The text below reproduces `PROTOCOL.md` from the code repository. Version 1.0 was committed at `f187a12` and the clarifications of version 1.1 at `0269f67`, both before any gate or test metric was computed. Note 1.2, at the end, was added after scoring; it documents differences between wording and implementation and changes no criterion.

S3.1 Analysis protocol (fixed before any test-set result)

Version 1.0, 4 November 2026. This file is committed to the repository before any test-set prediction is scored. Any later change is recorded at the end of this file with its date and reason; the original text is not edited.

S3.1.1 1. Question

Can an accept-or-abstain gate, calibrated on data held out *by chemistry*, keep the precision of a molecular model's accepted predictions at a stated target when the queries come from chemistry the model has not seen? And does combining an input-distance signal with a model-confidence signal rank errors better than either signal alone?

S3.1.2 2. Definitions fixed in advance

- **Correct prediction.** LogP: $|y - \hat{y}| \leq 0.5$ log units. Reaction yield: $|y - \hat{y}| \leq 10$ percentage points.
- **Coverage** is the share of queries the gate accepts. **Selective precision** is the share of accepted queries that are correct. **Selective error rate** = 1 − selective precision.
- **Target precisions:** t = 0.8 and t = 0.9. Risk level for the threshold guarantee: $\delta = 0.1$.

S3.1.3 3. Data (all public)

- **D1 LogP pool:** SangsterLogP main sheet (Zenodo 10.5281/zenodo.19387551, CC BY 4.0). Parsed with RDKit; unparsable SMILES dropped; duplicates by InChIKey merged (first record kept). Every compound whose InChIKey occurs in either external set below is removed from the pool.
- **E1, prospective external set:** the SangsterLogP "Prospective" sheet (327 compounds).

- **E2, fluorinated external set:** the 183 measured amide logP values of `blackthorn-ai/fluoricLogPpKa` (MIT), each fluorinated compound beside its non-fluorinated parent.
- **D3a Buchwald–Hartwig** (Ahneman et al. 2018; as redistributed in `SoftServeInc/yield-paper`, MIT): 3,955 reactions; the four published out-of-sample additive tests (sheets Test1–Test4, training rows 1–3057, 1–3055, 1–3058 and 1–3055 respectively) and random splits.
- **D3b Suzuki–Miyaura** (Perera et al. 2018; same redistribution): 5,760 reactions, target `Product_Yield_PCT_Area_UV`.

S3.1.4 4. Representations and base models

- Molecules: ECFP4 counts (radius 2, 2,048 bins) plus ten RDKit descriptors (MolWt, TPSA, NumHDonors, NumHAcceptors, NumRotatableBonds, RingCount, NumAromaticRings, FractionCSP3, HeavyAtomCount, NumHeteroatoms). Crippen MolLogP is deliberately excluded.
- Reactions: concatenated ECFP4 bit vectors (1,024 bits) of each variable component (BH: aryl halide, ligand, base, additive; SM: reactant 1, reactant 2, ligand, reagent, solvent). Component structures follow the cited redistribution.
- **M1 (primary):** random forest, 300 trees, max_features = 0.2, other settings at scikit-learn defaults. Spread u = standard deviation of the per-tree predictions.
- **M2 (secondary, robustness only):** ensemble of five multilayer perceptrons (512–128, ReLU, Adam, L2 = 1e-4, early stopping on 10% of the fit set). Spread u = standard deviation across members.
- Hyper-parameters are fixed here and are not tuned on any calibration or test fold.

S3.1.5 5. Splits (five seeds, 0–4)

For each dataset a pool is divided by a group label: Butina clusters of ECFP4 at Tanimoto distance 0.65 (D1), additive (D3a), ligand (D3b).

- **Held-out test H:** D1: 15% of clusters; D3a: the published TestN additives; D3b: two ligands per seed.
- **Held-out calibration C_h:** groups disjoint from H and from the fit set. D1: 15% of clusters; D3a: four additives drawn from the TestN training additives; D3b: two further ligands.
- **In-distribution part I** (all remaining groups) is split at random into the fit set F (70%), the random calibration fold C_r (15%) and the random test R (15%).
- The base model is trained on F only. Both gates (C_r-calibrated and C_h-calibrated) use the same model.
- **D1 external tests:** the D1 models are also scored on E1. A separate D1 model, trained after removing every compound with an sp^3 C–F bond (SMARTS `[CX4]F`) from the pool, is scored on E2; fluorinated compounds and parents are reported apart.

Held-out tests are therefore: D1 cluster, D1–E1, D1–E2 (D1 family); D3a Test1–Test4 and D3b ligand (D3 family).

S3.1.6 6. Signals

- **s1:** maximum Tanimoto similarity of the query fingerprint to F; **s5:** mean of the five largest.
- **u:** ensemble spread of the base model.
- **Conformal intervals (nominal 90%):** normalised split conformal, score $|y - \hat{y}| / (u + \varepsilon)$, $\varepsilon = 0.1 \times \text{median}(u)$ on the calibration fold.

S3.1.7 7. Gates

- Each calibration fold C is split at random into halves C_a and C_b.
- **Single-signal gates:** score s1, or −u.
- **Gate B (fused):** logistic regression of correctness on (s1, s5, log u), fitted on C_a. Its score is the predicted probability of a correct prediction.

- **Threshold by Learn-then-Test** on C_b, for every score: candidate thresholds are the score quantiles of C_b, tested in fixed sequence from the strictest that accepts at least 50 points of C_b, loosening one step at a time. A threshold passes when the binomial p-value $P(\mathrm{Bin}(n, t) \geq k) \leq \delta$, where n points are accepted and k are correct. The loosest threshold reached before the first failure is used. If none passes, the gate abstains on every query.
- **Gate A (conjunctive, no guarantee):** accept if $s1 \geq \tau_1$ and $u \leq \tau_2$; (τ_1, τ_2) from a 20×20 grid of quantiles of the whole C, chosen as the largest coverage whose empirical precision on C is at least t.

S3.1.8 8. Metrics

Per test set, model and seed: RMSE, MAE and R^2 ungated; error rate ungated; risk–coverage curves of selective error rate and selective RMSE for each score; AURC of the selective error rate (mean over all coverage levels k/n); coverage and achieved precision of every gate at t = 0.8 and 0.9; empirical coverage of the conformal intervals on all queries and on accepted queries. Results are averaged over seeds; 95% intervals come from 1,000 bootstrap resamples of test queries, pooled over seeds.

S3.1.9 9. Hypotheses and pass criteria (model M1)

- **H1, the gate helps where the model is weak.** On every held-out test, the selective error rate of the fused score at 30% coverage is lower than the ungated error rate, with the 95% bootstrap interval of the difference excluding zero.
- **H2, the calibration split matters.** Over the (held-out test × target) cells, Gate B calibrated on C_r falls short of the target by more than 0.05 in at least half of the cells; calibrated on C_h, it reaches at least $t - 0.05$ in at least three quarters of the cells (seed-averaged precision).
- **H3, the combination adds something.** The AURC of Gate B's score (C_h-fitted) is at least 10% lower, relatively, than the better single signal on at least two held-out tests in each of the D1 and D3 families.
- **H4, conformal coverage under shift.** Intervals calibrated on C_r cover at least 0.88 of R; cover less than 0.85 on at least one held-out test; and, on each held-out test where they fall below 0.85, cover at least 0.88 of the queries accepted by Gate B (C_h, t = 0.8).

Every hypothesis is reported whether or not it holds.

S3.1.10 10. Exclusions

No client, proprietary or confidential data, code or results are used. Biological context (gene expression, perturbation or cell-line shift) and generative models are out of scope.

S3.1.11 Changes after this version

1.1, 4 November 2026, before any gate or test metric was computed (base-model predictions were being generated; none had been scored). Three points the text above left open are fixed as follows.

- *A gate that abstains on every query* has no defined precision. Such a seed is left out of the seed average. If every seed abstains, the cell is reported as "abstained": it counts as meeting the target in H2 (no wrong prediction is accepted) and is flagged in the tables.
- *H4 on a test where Gate B accepts nothing* cannot be checked; it counts as a failure of H4 for that test.
- *Held-out tests used for H1–H4:* D1 cluster (logp/H), D1–E1 (logp/E1) and D1–E2 (logp_noCF/E2, all 183 compounds); D3a Test1–Test4 (bh_TestN/H) and D3b ligand (sm_ligand/H). The random test R of each task is the in-distribution reference.
- *Reduced training sets in D3a.* Holding out four calibration additives leaves about 1,640 reactions in F for each TestN, roughly half of the published training rows. Ungated accuracy is therefore not directly comparable with published Test1–Test4 results.

1.2, 4 November 2026, after scoring: documentation only. Nothing in sections 1–9 or in the pass criteria changes. Two independent reviews of the first manuscript draft found the points below.

- *Wording versus implementation.* Section 5 says the logP held-out test and calibration fold receive "15% of clusters". The code (`splits.group_holdout`) assigns whole clusters in random order until 15% of the *compounds* is reached. The manuscript describes the code.
- *Suzuki–Miyaura structures.* Section 4 says the component structures "follow the cited redistribution". They come from a table written for this repository (`src/aoa/data.py`), which omits spectator fragments: the internal standard, HCl and water of hydration.
- *Count in note 1.1.* Note 1.1 announces three points and lists four.
- *Bootstrap correction.* The first version of `boot_h1` resampled each seed independently. For test sets that are identical in every seed (E1, E2, BH Test1–Test4), that makes the intervals too narrow. The corrected code uses the same resample for every seed in those sets. No H1 verdict changes; the E2 interval widens from [−0.064, +0.029] to [−0.098, +0.067].
- *Timing.* The whole study ran in one working session on 4 November 2026. The commit history shows the order: the empty repository, then protocol 1.0, then 1.1, then the code, then the results. Base-model predictions were generated with the committed code shortly before that code was committed; none were scored before `feadf9a`.
- *Exploratory analyses.* The following are not in the protocol and are reported as exploratory (`scripts/exploratory.py`):
  - per-seed target attainment;
  - pooled precision in the operating-limits cards;
  - conformal intervals calibrated on C_h;
  - the E2 signal diagnostics.

S4 Software environment

Python 3.12.14; RDKit 2026.09.1; scikit-learn 1.9.1; NumPy 2.5.4; SciPy 1.18.1; pandas 3.1.0; matplotlib 3.11.2. The full pipeline (`fetch_data.py`, `run_predictions.py`, `analyse.py`, `make_figures.py`, `make_tables.py`) runs in about 40 minutes on an 18-core laptop CPU. Input digests (SHA-256) are listed in `PROVENANCE.md` and checked by `fetch_data.py`.